Tobias Stephan

Assistant Professor of Structural Geology | Department of Geology, Lakehead University, Thunder Bay, ON, Canada

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1 EDUCATION

2019/03 **Doctor of Philosophy (PhD)** in "Geology"
Technische Universität Bergakademie Freiberg, Germany

2013/09 **Master of Science (MSc)** in "Geosciences"
Technische Universität Bergakademie Freiberg, Germany

2010/09 **Bachelor of Science (BSc)** in "Geology and Mineralogy"
Technische Universität Bergakademie Freiberg, Germany

2 PROFESSIONAL EXPERIENCE

since 2025/12 **Assistant Professor**
Lakehead University, Department of Geology, Thunder Bay, ON, Canada

2023/04–2023/11 **Postdoctoral associate / Lecturer**
Lakehead University, Department of Geology, Thunder Bay, ON, Canada

2020/12–2022/11 **DFG Research Fellow**
University of Calgary, Geo- and Thermochronology Research Group, Department of Geoscience, Calgary, AB, Canada

2020/03–2020/11 **Postdoctoral associate**
Friedrich-Alexander-Universität Erlangen-Nürnberg, Geozentrum Nordbayern, Erlangen, Germany

2019/09–2019/12 **Research assistant**
Technische Universität Bergakademie Freiberg, Institute for Computer Sciences, Freiberg, Germany

2014/01–2014/06 **Geologist**
Beak Consultants GmbH (Germany / Tanzania)

3 TEACHING EXPERIENCE

since 2026 Metamorphic Petrology (GEOL-3217)
Lakehead University, Department of Geology, Thunder Bay, ON, Canada
Course level: undergraduate | lecture hours per week: 3
course syllabus

since 2025 Geology Case Studies (GEOL–4313)
Lakehead University, Department of Geology, Thunder Bay, ON, Canada
Course level: undergraduate | lecture hours per week: 3
course syllabus

since 2024 Structural Geology & Tectonics (GEOL-3310)
Lakehead University, Department of Geology, Thunder Bay, ON, Canada
Course level: undergraduate | lecture hours per week: 3
course syllabus

4 PUBLICATIONS

4.1 PEER-REVIEWED ARTICLES

- Stephan, T., Pinto, T. F., and Enkelmann, E. (submitted): "Cluster Analysis of Inverse Thermochronology Models. Part 1: Statistical Method and Implementation ". *Geosphere*

- Pinto, T. F., Stephan, T., and Enkelmann, E. (submitted): "Cluster analysis of Inverse Thermochronology Models. Part 2: Geological Applications". *Geosphere*

17 Pinto, T. F., Enkelmann, E., Stephan, T., Beranek, L., Padgett, J., Terlaky, V., and Qadri, T. (accepted): "Long-term effect of inherited basement structures on the basin evolution of northwestern Canada". *GSA Bulletin*

16 Stephan, T., Phillips, N., Tiitto, H., Perez, A., Nwakanma, M., and Hollings, P. (2025): "Going with the flow — Changes of Vorticity Controls Gold Enrichment in Archean Shear Zones (Shebandowan Greenstone Belt, Superior Province, Canada)". *Journal of Structural Geology* 201. doi: 10.1016/j.jsg.2025.105542.

15 Stephan, T., and Enkelmann, E. (2025): "All Aligned on the Western Front of North America? Analyzing the Stress Field in the Northern Cordillera". *Tectonics* 44(9). doi: 10.1029/2025TC009014.

14 Padgett, J., Enkelmann, E., Kellett, D., Moynihan, D., and Stephan, T. (2025): "Cenozoic faulting in the Upper Hyland River Valley, Southeastern Yukon: A thermochronological perspective". *Canadian Journal of Earth Sciences* 62(5): 1002-1022. doi: 10.1139/cjes-2024-0147.

13 Schaeben, H., Kroner, U., and Stephan, T. (2024): "Mathematical Fundamentals of Spherical Kinematics of Plate Tectonics in Terms of Quaternions". *Mathematical Models and Methods in Applied Sciences* 47(6): 4469–4496. doi: 10.1002/mma.9823

12 Stephan, T., Enkelmann, E., and Kroner, U. (2023): "Analyzing the horizontal orientation of the crustal stress adjacent to plate boundaries". *Scientific Reports* 13:15590. doi: 10.1038/s41598-023-42433-2.

11 Járóka, T., Pfänder, J. A., Seifert, T., Hauff, F., Sperner, B., Staude, S., Stephan, T., and Schulz, B. (2023): "Age and petrogenesis of Ni-Cu-(PGE) sulfide-bearing gabbroic intrusions in the Lausitz Block, northern Bohemian Massif (Germany/Czech Republic)". *Lithos* 444–445:107090. doi: 10.1016/j.lithos.2023.107090

10 Kroner, U., Romer, R. L., and Stephan, T. (2023): "Die Rekonstruktion von relativen Plattenbewegungen aus dem paläozoischen Deformationsmuster der kontinentalen Kruste". *Zeitschrift der Deutschen Gesellschaft für Geowissenschaften (J. Appl. Reg. Geol.)*. doi: 10.1127/zdgg/2023/0365

9 Köhler, S., Duschl, F., Fazlikhani, H., Koehn, D., Stephan, T., and Stollhofen, H. (2022): "Reconstruction of cyclic Mesozoic-Cenozoic stress development in SE Germany using fault-slip and stylolite inversion". *Geological Magazine* 159 (11–12): 2323–2345. doi: 10.1017/S0016756822000656

8 Kroner, U., Stephan, T., and Romer, R. L. (2022): "Paleozoic orogenies and relative plate motions at the sutures of the Iapetus-Rheic Ocean". In Y. D. Kuiper, J. B. Murphy, R. D. Nance, R. A. Strachan, and M. D. Thompson (Eds.), *New Developments in the Appalachian-Caledonian-Variscan Orogen*. Geological Society of America. doi: 10.1130/2021.2554(01)

7 Schaeben, H., Kroner, U., and Stephan, T. (2021): "Euler Poles of Tectonic Plates". In B. S. Daza Sagar, Q. Cheng, J. McKinley, and F. Agterberg (Eds.), *Encyclopedia of Mathematical Geosciences. Encyclopedia of Earth Sciences Series. Springer Nature* Switzerland AG 2021. doi: 10.1007/978-3-030-26050-7_435-1

6 Caracciolo, L., Ravidà, D. C. G., Chew, D., Janßen, M., Lünsdorf, N. K., Heins, W. A., Stephan, T., and Stollhofen, H. (2021): "Reconstructing environmental signals across the Permian-Triassic boundary in the SE Germanic Basin: A Quantitative Provenance Analysis (QPA) approach". *Global and Planetary Change*, 206:103631. doi: 10.1016/j.gloplacha.2021.103631

5 Kroner, U., Stephan, T., Romer, R. L., and Roscher, M. (2020): "Paleozoic plate kinematics during the Pannotia–Pangaea supercontinent cycle". *Geological Society, London, Special Publications* 503, SP503-2020-15. doi: 10.1144/SP503-2020-15

4 Stephan, T., Kroner, U., Romer, R. L., and Rösel, D. (2019): "From a bipartite Gondwana shelf to the arcuate Variscan belt: The Early Paleozoic evolution of northern Peri-Gondwana". *Earth-Science Reviews* 192: 491–512. doi: 10.1016/j.earscirev.2019.03.012

3 Heinicke, J., Stephan, T., Alexandrakis, C., Buske, S., and Gaupp, R. (2019): "Alteration as possible cause for transition from brittle failure to aseismic slip: the case of the NW-Bohemia / Vogtland earthquake swarm region". *Journal of Geodynamics* 124: 79–92. doi: 10.1016/j.jog.2019.01.010

2 Stephan, T., Kroner, U., and Romer, R. L. (2018): "The pre-orogenic detrital zircon record of the Peri-Gondwanan crust". *Geological Magazine* 156(2): 281–307. doi: 10.1017/s0016756818000031.

1 Stephan, T., Kroner, U., Hahn, T., Hallas, P., and Heuse, T. (2016): "Fold / cleavage relationships as indicator for late Variscan sinistral transpression at the Rheno-Hercynian–Saxo-Thuringian boundary zone, Central European Variscides". *Tectonophysics* 681: 250–262. doi: 10.1016/j.tecto.2016.03.005

4.2 OTHER ACADEMIC ARTICLES

Book Legler, C., Barth, A., Knobloch, A., Mruma, A. H., Myumbilwa Y., Magigita, M., Msechu, M., Ngole, T., Stanek, K. P., Boniface, N., Kagya, M., Manya, S., Berndt, T., Stahl, M., Gebremichael, M., Dickmayer, E., Repper, C., Falk, D., and Stephan, T. (2015): "Explanatory Notes for the Minerogenic Map of Tanzania 1:1,5 M.", *Geological Survey of Tanzania*. ISBN: 978-9987-477-94-4

5 SOFTWARE DEVELOPMENTS

`tectonicr` Free and open-source R package for modeling and analyzing the direction of the maximum horizontal stress using relative plate motion.
doi: 10.32614/CRAN.package.tectonicr | `https://tobiste.github.io/tectonicr/`

`structr` Free and open-source R package for analyzing and visualizing orientation data for structural geology.
`https://tobiste.github.io/structr/`

`geocreep` Estimate uncertainties in strain rates from flow laws, grain-size piezometry, and fugacity computations.
`https://github.com/tobiste/geocreep/`

`geoprofiler` Creates Swath profiles and Distance vs X plots by measuring the accurate distances parallel and perpendicular to user-defined lines.
doi: 10.32614/CRAN.package.geoprofiler | `https://tobiste.github.io/geoprofiler/`

`thermoclustr` Clustering, advanced statistics and densities of thermochronology cooling paths from inverse models.
`https://github.com/tobiste/thermoclustr/`

`laftr` Free and open-source R package to calculate the ages from LA-ICP-MS based fission track dating using the zeta approach.
`https://tobiste.github.io/laftr/`

6 FUNDING AND GRANTS

2026–2031 *Natural Sciences and Engineering Research Council of Canada (NSERC)* Discovery Grant (**PI**)
Title: "Kinematics and Strain Geometry of Archean Tectonics — A New Tool for Predicting Gold Mineralization in the Canadian Shield" | Amount: CA$ 183,000

2026–2031 *Natural Sciences and Engineering Research Council of Canada (NSERC)* Alliance Grant
Title: "Tectonostratigraphy and structural evolution of the Great Bear project, NW Ontario" (Co-PI) | Amount: CA$ 800,000

2026–2028 *Natural Sciences and Engineering Research Council of Canada (NSERC)* Alliance Grant
Title: "Alteration chemistry of the Tower Mountain gold deposit" (Co-PI) | Amount: CA$ 100,000

2020–2022 *German Research Foundation (DFG)* Research Fellowship (**PI**)
Title: "Developing a statistical approach to analyze large paired geo-thermochronological datasets with an application to the Canadian Cordilleras" | Amount: € 170,000

7 AWARDS

2026 CTG David Elliott Best Paper Award (2026) for *"All Aligned on the Western Front of North America? Analyzing the Stress Field in the Northern Cordillera"* (Tectonics, 2025)

8 INVITED PRESENTATIONS

2026/10/14 Canadian Tectonics Group Seminars

2022/10/18 Lakehead University, Geology Seminar Series

2022/05/11 Geological Survey Canada, McConnell Club Talks

9 PROFESSIONAL SERVICES AND MEMBERSHIPS

Memberships The Geological Association of Canada (GAC), Canadian Tectonics Group (CTG)

Reviewer for journals Geology, Gondwana Research, Terra Nova, Geological Society of America, Scientific Reports, GSA Bulletin, Proceedings of the Geologists' Association, Basin Research, Lithosphere, Discover Geoscience, Tectonics

Reviewer for grant proposals National Science Center, Poland

Committee board member Secretary of the Canadian Tectonics Group (GAC subdivision) since 2026

Session Chair *GAC-MAC-PEG 2024* (Brandon, MN, Canada): "It's our fault! Geological and geophysical insights into fault and shear zone processes"

October 16, 2026